RetailIQ

AI-Powered Sales Amount Prediction System for Retail & E-Commerce

Overview

Problem Statement

Retail businesses struggle with manual checkout value estimation and static averages, leading to poor inventory planning and lost sales margins. RetailIQ uses machine learning to accurately estimate transaction sales amounts from historical data.

Business Objectives & Success Criteria

1. Improve sales forecasting accuracy at checkout. Optimize pricing, promotions, and discount strategies.. Support inventory planning and increase revenue with data driven decisions.

R^2 SCORE
High Variance Explained (Low error across test split)

MAE
Mean Absolute Error (Average error per transaction)

RMSE
Root Mean Squared Error (Penalizes large anomalies)

AI Methodology

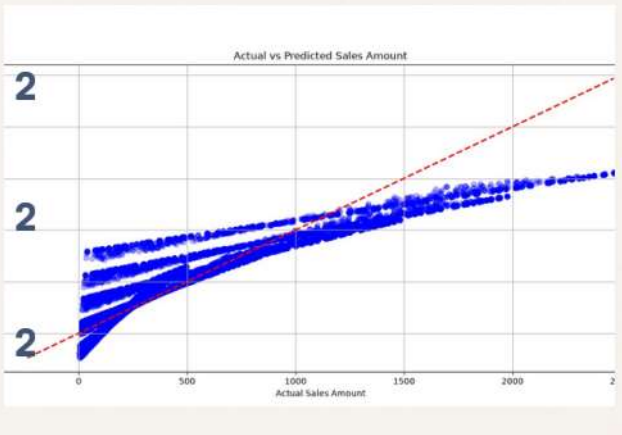
Data & Model Pipeline

- Model Approach: Supervised Machine Learning using Multiple Linear Regression.
- Categorical Features: sales_channel, payment_method, and region (encoded using One-Hot Encoding).
- Date Features: Extracted year, month, day, and day_of_week.
- Target Variable: sales_amount (checkout transaction value).
- Data Leakage Safeguard: Excluded gross_sales and discount_amount to prevent artificial accuracy inflation.
- Data Split Strategy: 60% Training / 40% Testing hold-out split (random_state=42 for reproducibility).

Actual vs Predicted Sales Plot

Model Results & Insights

- High Predictive Accuracy: Data points closely follow the 45° dashed red reference line, confirming consistent sales estimates across standard transaction ranges.
- Feature Clustering: Visible tiered bands in the scatter plot capture real-world groupings across product categories, item quantities, and sales channels.
- Model Serialization: The trained regression model and its feature columns are saved via joblib (sales_model.pkl and model_features.pkl) for instant, reproducible predictions without retraining.



Tools & Technologies



Community Benefit & Chatbot Assistant

- Technical Stack: Python, Pandas, NumPy, Scikit-Learn, Matplotlib, GitHub, and Canva.
- SalesBoost Assistant: An interactive chatbot that provides sales analysis, performance insights, and decision support.

Community Benefit

- Economic Support: Helps local retail businesses remain financially sustainable, competitive, and operationally efficient.
- Consumer Impact: Promotes price stability and ensures reliable, continuous community access to everyday essential goods.

Constraints & Risks

- Constraints: Predictions are limited to the variables available in the dataset and estimate transaction value rather than inventory stock levels.
- Risks & Mitigation: Poor-quality inputs and missing data risks are minimized through systematic data cleaning, validation, and one-hot encoding pipelines.